



SOHCAHTOA: Word Problems on Right Triangles

Trigonometry Worksheet · Grade 9-10 · numberbender.com

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Learning Objectives

- Apply the SOHCAHTOA ratios (sine, cosine, tangent) to find missing sides of a right triangle in real-world contexts
- Distinguish between angle of elevation and angle of depression, and correctly set up the corresponding trigonometric equation
- Use inverse trigonometric functions to determine unknown angles in a right triangle given two known side lengths

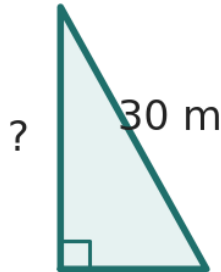
For each problem, sketch a right triangle (or use the diagram provided), label the known and unknown sides, select the correct trigonometric ratio using SOHCAHTOA, and solve; round all answers to two decimal places unless stated otherwise.

1. From a point 80 meters from the base of a building, the angle of elevation to the top of the building is 42° . Find the height of the building to the nearest meter.

$$\tan(42^\circ) = \frac{h}{80}$$

Answer: _____

2. A 30-meter ladder leans against a vertical wall, making a 65° angle with the ground. Find the height h at which the ladder touches the wall.



Answer: _____

3. From the top of a cliff 96 meters high, a ship is spotted at an angle of depression of 24° . Find the horizontal distance d from the base of the cliff to the ship.

$$\tan(24^\circ) = \frac{96}{d}$$

Answer: _____



4. A kite is attached to a taut 120-meter string that makes a 38° angle with the horizontal ground. Find the height h of the kite above the ground.

$$\sin(38^\circ) = \frac{h}{120}$$

Answer: _____

5. A tree casts a shadow 45 feet long on level ground. If the angle of elevation of the sun is 53° , find the height of the tree.

$$\tan(53^\circ) = \frac{h}{45}$$

Answer: _____

6. An airplane flying at an altitude of 2,500 meters spots an airport runway at an angle of depression of 32° . Find the horizontal distance d between the airplane and the runway.

$$\tan(32^\circ) = \frac{2500}{d}$$

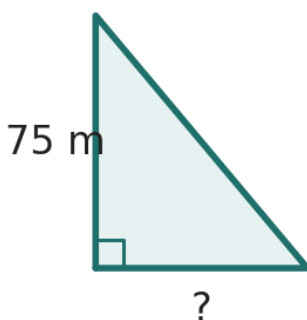
Answer: _____

7. A right triangle has legs measuring 9 cm and 40 cm. Find the measure of the acute angle θ that is opposite the 9-cm leg. Round to the nearest hundredth of a degree.

$$\tan(\theta) = \frac{9}{40}$$

Answer: _____

8. A person standing at a horizontal distance d from a 75-meter-tall lighthouse observes the top at an angle of elevation of 50° . The diagram shows the right triangle formed. Find the distance d .



Answer: _____

9. From a window 20 meters above the ground, a person observes a car parked on the street at an angle of depression of 35° . Find the horizontal distance d from the base of the building to the car.

$$\tan(35^\circ) = \frac{20}{d}$$

Answer: _____





Remind students that the angle of depression from an elevated observer always equals the angle of elevation from the object below, because they are alternate interior angles formed by a transversal cutting two parallel horizontal lines — drawing both angles on the same diagram reinforces this relationship.

Solutions

1. From a point 80 meters from the base of a building, the angle of elevation to the top of the building is 42° . Find the height of the building to the nearest meter.

$$\tan(42^\circ) = \frac{h}{80}$$

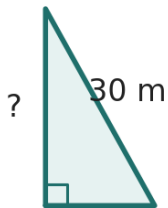
→ Draw a right triangle: the 80-m horizontal distance is the adjacent side and h is the opposite side, with the 42° angle at the observer's feet

→ Use tangent: $\tan(42^\circ) = \text{opposite} / \text{adjacent} = h / 80$

→ Solve: $h = 80 \times \tan(42^\circ) \approx 80 \times 0.9004 \approx 72 \text{ m}$

Answer: $h = 80 \cdot \tan(42^\circ) \approx 72 \text{ m}$

2. A 30-meter ladder leans against a vertical wall, making a 65° angle with the ground. Find the height h at which the ladder touches the wall.



→ The ladder is the hypotenuse (30 m) and h is the side opposite the 65° angle at the ground

→ Use sine: $\sin(65^\circ) = \text{opposite} / \text{hypotenuse} = h / 30$

→ Solve: $h = 30 \times \sin(65^\circ) \approx 30 \times 0.9063 \approx 27.19 \text{ m}$

Answer: $h = 30 \cdot \sin(65^\circ) \approx 27.19 \text{ m}$

3. From the top of a cliff 96 meters high, a ship is spotted at an angle of depression of 24° . Find the horizontal distance d from the base of the cliff to the ship.

$$\tan(24^\circ) = \frac{96}{d}$$

→ The angle of depression (24°) from the cliff top equals the angle of elevation from the ship — alternate interior angles with the horizontal line

→ The cliff height (96 m) is the opposite side and d is the adjacent side

→ Set up: $\tan(24^\circ) = 96 / d \rightarrow d = 96 / \tan(24^\circ) \approx 96 / 0.4452 \approx 215.60 \text{ m}$

Answer: $d = \frac{96}{\tan(24^\circ)} \approx 215.60 \text{ m}$



4. A kite is attached to a taut 120-meter string that makes a 38° angle with the horizontal ground. Find the height h of the kite above the ground.

$$\sin(38^\circ) = \frac{h}{120}$$

→ The string is the hypotenuse (120 m) and h is the vertical height opposite the 38° angle

→ Use sine: $\sin(38^\circ) = \text{opposite} / \text{hypotenuse} = h / 120$

→ Solve: $h = 120 \times \sin(38^\circ) \approx 120 \times 0.6157 \approx 73.88 \text{ m}$

Answer: $h = 120 \cdot \sin(38^\circ) \approx 73.88 \text{ m}$

5. A tree casts a shadow 45 feet long on level ground. If the angle of elevation of the sun is 53° , find the height of the tree.

$$\tan(53^\circ) = \frac{h}{45}$$

→ The shadow (45 ft) is the adjacent side and the tree height h is the opposite side relative to the 53° sun angle

→ Use tangent: $\tan(53^\circ) = h / 45$

→ Solve: $h = 45 \times \tan(53^\circ) \approx 45 \times 1.3270 \approx 59.72 \text{ ft}$

Answer: $h = 45 \cdot \tan(53^\circ) \approx 59.72 \text{ ft}$

6. An airplane flying at an altitude of 2,500 meters spots an airport runway at an angle of depression of 32° . Find the horizontal distance d between the airplane and the runway.

$$\tan(32^\circ) = \frac{2500}{d}$$

→ The angle of depression (32°) from the plane equals the angle of elevation from the runway — alternate interior angles

→ Set up: $\tan(32^\circ) = \text{altitude} / \text{horizontal distance} = 2500 / d$

→ Solve: $d = 2500 / \tan(32^\circ) \approx 2500 / 0.6249 \approx 4000.54 \text{ m}$

Answer: $d = \frac{2500}{\tan(32^\circ)} \approx 4000.54 \text{ m}$

7. A right triangle has legs measuring 9 cm and 40 cm. Find the measure of the acute angle θ that is opposite the 9-cm leg. Round to the nearest hundredth of a degree.

$$\tan(\theta) = \frac{9}{40}$$

→ The angle θ is opposite the 9-cm leg and adjacent to the 40-cm leg

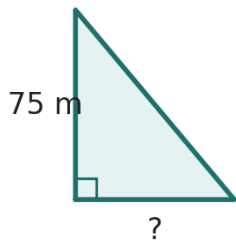
→ Use tangent: $\tan(\theta) = \text{opposite} / \text{adjacent} = 9/40 = 0.225$

→ Apply the inverse tangent: $\theta = \tan^{-1}(0.225) \approx 12.68^\circ$

Answer: $\theta = \tan^{-1}\left(\frac{9}{40}\right) \approx 12.68^\circ$



8. A person standing at a horizontal distance d from a 75-meter-tall lighthouse observes the top at an angle of elevation of 50° . The diagram shows the right triangle formed. Find the distance d .



- The lighthouse height (75 m) is the opposite side and d is the adjacent side, with the 50° angle at the observer
- Use tangent: $\tan(50^\circ) = \text{opposite} / \text{adjacent} = 75 / d$
- Solve: $d = 75 / \tan(50^\circ) \approx 75 / 1.1918 \approx 62.97 \text{ m}$

Answer: $d = \frac{75}{\tan(50^\circ)} \approx 62.97 \text{ m}$

9. From a window 20 meters above the ground, a person observes a car parked on the street at an angle of depression of 35° . Find the horizontal distance d from the base of the building to the car.

$$\tan(35^\circ) = \frac{20}{d}$$

- The angle of depression (35°) from the window equals the angle of elevation from the car
- The window height (20 m) is the opposite side and d is the adjacent side
- Set up: $\tan(35^\circ) = 20 / d \rightarrow d = 20 / \tan(35^\circ) \approx 20 / 0.7002 \approx 28.56 \text{ m}$

Answer: $d = \frac{20}{\tan(35^\circ)} \approx 28.56 \text{ m}$

